MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746268
TIMING-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements of Timing-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-G-48411 | Circuit Card Assemblies, General Specification for |

DRAWINGS

| | |
|---|---|
| 11746768 | Circuit Card Assembly, Timing-1 |

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition.- This assembly is one (1) of three (3) single ended printed circuit boards consisting of a network oscillators, counters, comparators, flip-flops, inverts and gates necessary to provide all timing pulses to initiate the various maneuvers performed by the aircraft. The network also generates the analog times to be used in the analog computer section of the STG.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746268.



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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with a 5 VDC power supply.

3.2.1.2 Timing-1 Network.- The outputs for the associated functions shall be as shown in Table I & II when the respective inputs are applied.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-G-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746268 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-G-48411, Drawing 11746268 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-G-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-G-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-G-48411.



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TABLE I

| PINS / FUNCTION | INPUT 9 | 10 | 8 | 6 | 28 | 24 | 30 | 32 | 41 | 22 | OUTPUT 2 | 3 | 21 | 18 | 37 | 5 | 39 | 34 | 40 | 19 | 20 | 38 | 26 | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4K Hz Clock Test | | | | | | | | | | | | | | | | ⊓ | | | | | | | | f=3.93±0.02KHz |
| 15 Hz Clock Test | | | | | | | | | | | ⊓ | | | | | | | | | | | | | f=15.38±0.1Hz |
| 60Hz Gated Clock Test | 0<br>1 | | | | | | | | 1 | | | | 1<br>⊓ | | | | | | | | | | | f=61.5±0.3Hz<br>Pulse Duration of 58±2 sec. |
| 15 Hz Gated Clock Test | 0<br>1 | | | | | | | | 1 | | | | 1<br>⊓ | | | | | | | | | | | f=15.38±0.1Hz<br>Pulse Duration of 58±2 sec. |
| Sin Ψ Inhibit Test | | | 0<br>1<br>1<br>1 | 1<br>1<br>0 | | | | | | | | | | | | | | | | 1<br>0<br>1<br>1 | | | | |
| $Q_1, Q_2, Q_3, Q_4$ Test | 0<br>1<br>0<br>1<br>0<br>1<br>0<br>1 | | | | | | | | 1<br>1<br>1<br>1 | 0 | | | ⊓ | | | ⊓ | | ⊓ | | | | ⊓ | | Pulse Duration 4.16±0.1 sec.<br>Pulse Duration 8.32±0.1 sec.<br>Pulse Duration 16.64±0.1 sec.<br>Pulse Duration 33.2±0.1 sec. |
| Stop Clock Test | 0<br>1 | | | | | | | | 1 | | | | | | | | | | | | | ⊓ | | Pulse Duration 58.16±0.1 sec. |
| 1 Hz Gated Clock Test | 0<br>1 | | | | | | | | 1 | | | | | | ⊓ | | | | | | | | | f=1.04±0.1Hz |

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TABLE II

| FUNCTION \ PINS | INPUT 9 | 10 | 8 | 6 | 28 | 24 | 30 | 32 | 41 | 22 | OUTPUT 2 | 3 | 21 | 18 | 37 | 5 | 39 | 34 | 40 | 19 | 20 | 38 | 26 | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Turn Gate and Turn Clock Pulse Test | 0 | 1 | 0 | 0 | | | | 1 | 1 | 0 | | | | | | | | | | | | | | Logic 0<br>Pulse Duration 4.16±0.1 sec.<br>f=61.5±0.3Hz<br>Pulse Duration 4.16±0.1 sec. |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | 0 | 1 | | | | | | 1 | 0 | | | | | | | | | | | | | | Pulse Duration 8.32±0.1 sec.<br>f=61.5±0.3Hz<br>Pulse Duration 8.32±0.1 sec. |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | 0 | 0 | 1 | | | | 1 | 1 | 0 | | | | | | | | | | | | | | Pulse Duration 16.64±0.1 sec.<br>f=61.5±0.3Hz<br>Pulse Duration 16.64±0.1 sec. |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | | | 0 | | | | | 1 | 0 | | | | | | | | 0 | | | 1 | | | |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | 1 | | | 0 | | 1 | | 1 | 0 | | | | | | | | | | | | | | Pulse Duration 4.16±0.1 sec.<br>f=61.5±0.3Hz |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | | | | | 1 | 0 | | 1 | 0 | | | | | | | | | | | | | | "0" Pulse Duration of 4.16±0.1 sec.<br>"1" Pulse Duration of 12.48±0.1 sec.<br>"1" Pulse Duration of 4.16±0.1 sec.<br>"0" Pulse Duration of 12.48±0.1 sec. |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | 0 | | | | 1 | 0 | | | 1 | 0 | | | | | | | | | | | | | | "0" Pulse Duration of 4.16±0.1 sec.<br>"1" Pulse Duration of 29.04±0.1 sec.<br>"1" Pulse Duration of 4.16±0.1 sec.<br>"0" Pulse Duration of 29.04±0.1 sec. |
| | 1 | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | 0 | | | | | | | | | | 1 | | | | | | |



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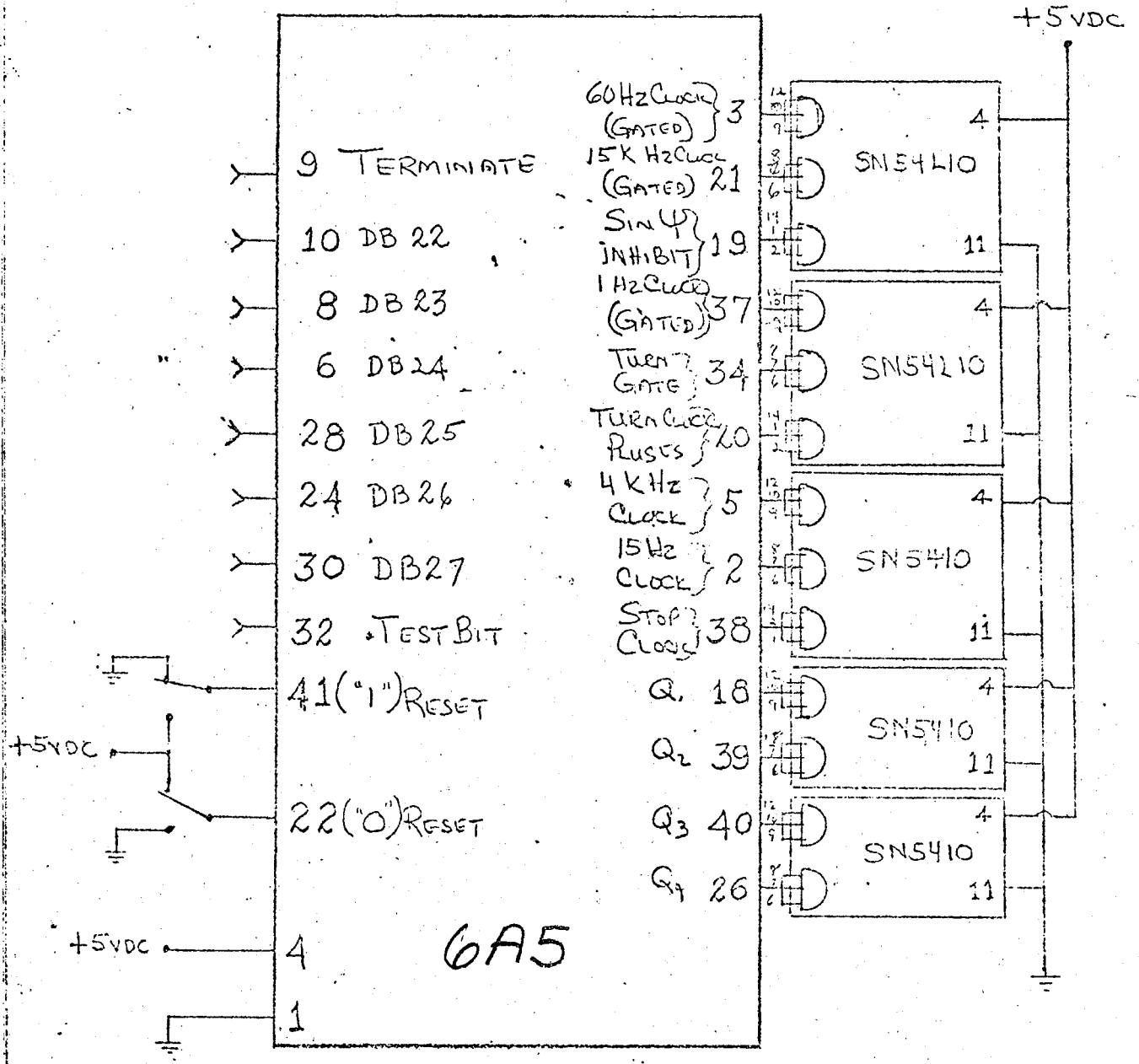



FIGURE 1



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4. QUALITY ASSURANCE PROVISIONS

4.1 General.-

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-G-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-G-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-G-48411.

4.2.1 Test equipment. (see Figure 1)

4.2.1.1 Power supply.- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS or 100 mV peak to peak.

4.2.1.2 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.3 Dummy loads.- The dummy loads shall be two quadruple 3-input positive nand gates or the equivalent to SN5410 and SN54L10.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these test shall be defined as:

a. Temperature between 60°F and 90°F
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All requirements shall be met with a power supply of +5.00 ± 0.05 VDC.

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.



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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746268.

6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applicable to this specification.

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Preparing Activity:

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Project No. 1240-A600

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